

Claim dependency graph

Repository source: proof/CLAIM_DEPENDENCIES.md

Candidate theorem

For every three-dimensional convex body of constant width d ,

$$\text{Vol}(K) \geq \frac{\pi}{1914} \left(259 - 27 \sqrt{\frac{15183}{17303}} \right) d^3 \approx 0.3836027047 d^3.$$

The integrated proof is in:

`paper/degree3_spectral_refinement.tex`

The repository does **not** yet label the statement a peer-reviewed, independently verified, or certified theorem.

Dependencies

1. Nishioka support-function framework

- h is odd after subtracting $d/2$ from the support function.
- A translation removes the degree-one harmonic component.
- $A_h = \nabla^2 h + h I$ satisfies $-dI/2 \leq A_h \leq dI/2$.
- $\text{Vol}(K) = \pi d^3/6 - d \cdot E(h)/2$.
- Smooth same-width bodies approximate arbitrary constant-width bodies in Hausdorff distance, with volume convergence.

2. Refined spectral split

- Write $h = h_3 + h_{\geq 5}$.
- Tensor-valued harmonic orthogonality gives $\|A_h\|_2^2 = \|A_{h_3}\|_2^2 + \|A_{h_{\geq 5}}\|_2^2$.
- Degree 3 has coefficient $1/22$; all odd degrees at least 5 have coefficient at most $1/58$.
- Therefore $E(h) \leq \|A_h\|_2^2/58 + (9/319) \|A_{h_3}\|_2^2$.
- Complete proof: `proof/BRIDGE_LEMMAS.md`, Sections 2–3.

3. New degree-3 determinant identity

- Represent $h_3(u) = T[u, u, u]$ by a symmetric trace-free rank-3 tensor.
- For $C = A_{h_3}$ and $B_{ij} = T_{ikl} T_{jkl}$, with $B_0 = B - \text{tr}(B)I/3$, prove

$$\|C\|_2^2 = \frac{176\pi}{7} \tau,$$

$$\|4 \det C\|_2^2 = \frac{\pi}{1001} (555264\tau^2 - 2265600\|B_0\|_F^2).$$

- Hence

$$\|4 \det C\|_2^2 \leq \frac{15183}{17303\pi} \|C\|_2^4.$$

- Human-readable derivation: `proof/DEGREE3_TENSOR_IDENTITY.md`.
- Exact symbolic verification: `python/verify_degree3_identity.py` and `python/verify_degree3_identity_alt.py`.
- Exact contraction-graph count: `python/verify_pairing_counts.py`, with a matching base-R implementation in `R/verify_pairing_counts.R`.

4. Operator/nuclear-norm bridge

- For a symmetric 2×2 matrix, $\|C\|_*^2 = \|C\|_F^2 + 2|\det C|$.
 - Cauchy–Schwarz and the determinant estimate bound `integral` $\|C\|_*^2$.
 - Tensor-harmonic orthogonality and $\|A_h(u)\|_{op} \leq d/2$ bound $\|C\|_2^2$.
 - Complete proof: `proof/BRIDGE_LEMMAS.md`, Sections 4–5.
5. **Final arithmetic**
- Insert $\|A_h\|_2^2 \leq 2\pi d^2$ and the refined degree-3 bound into the spectral split.
 - Apply Nishioka’s volume identity.
 - Complete proof: `proof/BRIDGE_LEMMAS.md`, Section 6.
 - Exact arithmetic check: `python/verify_bound_arithmetic.py`.
6. **Regularity passage**
- Prove the smooth case first.
 - Pass to arbitrary constant-width bodies using same-width smooth approximation and volume continuity.
 - Complete proof: `proof/BRIDGE_LEMMAS.md`, Section 7.
7. **Integrated publication draft**
- All dependencies above are assembled in `paper/degree3_spectral_refinement.tex`.
 - The compiled PDF is `paper/degree3_spectral_refinement.pdf`.
 - The internal line-by-line audit is `proof/PROOF_AUDIT.md`.
 - Build instructions are in `paper/README.md` and `build_paper.sh`.
8. **Still external to the proof package**
- Independent mathematical review.
 - A systematic literature/novelty review.
 - Verification of HYRA’s stronger public claims.